

MÉTODOS MATEMÁTICOS EN BIOINGENIERÍA

TEST # 4

DUVAN HENAO

A bacterial culture is grown in a fixed environment with an initial population of $N_0 = 500$ cells and an estimated carrying capacity of $K = 10000$ cells. The intrinsic growth rate (r) is of 0.4 per hour.

P1. Using the logistic equation:

$$N(t) = \frac{K}{1 + \left(\frac{K}{N_0} - 1\right) e^{-rt}}$$

find the population after $t = 1$ hour.

P2. Taking the Python code provided as a starting point, find the value of t when $N(t) = 8000$. Explain, line by line, how the code works.

P3. Suppose that in a different bacterial population, with $N_0 = 500$ cells and carrying capacity of $K = 10000$ cells, the population after 1 hour ($t = 1$) has increased to 3000 cells. Adapt the code in Python so as to identify the intrinsic growth rate r .